Choose Informative Questions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Teach question usefulness for an explicit goal.

Supply the full card table, separate current game tasks from retained Week 14 review and assess the ten-card comparison consistently.

Keep one habit: Require the current population, both branch counts, criterion and a limit on the conclusion.

MAIN LESSON · SECTION 1

Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

Learning goal and precision

Primary target: Compare questions by their largest remaining group

Teach a two-session worst-case question investigation. Supply the complete eight-card table, count Yes/No groups for each candidate, then restrict to four mystery cards and recount. Introduce tuple positions and Boolean counting only if using the optional Python

audit. Retain earlier tree-fitting content as Week 14 review; do not treat it as the current learning target or independent assessment.

By the end, I can choose a question that separates possibilities rather than repeating a constant clue.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- Case exposure check: this source example appears in ai-2/14, ai-2/15, ai-4/11. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.
- Printed lesson and workbook
- Eight catalogue cards M1–M4,S1–S4 with subject, binding and language shown; a separate ten-card independent count brief
- Two **branch** areas labelled Yes and No; a count record with the largest remaining group
- Optional Python 3 editor with no packages required

Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

Look for: Fit a **root** question and **leaf prediction**s from **training** examples

Reference example, not the only acceptable wording: The **root** asks near, with **branch**es 0 and 1. Leaves predict clear for 0 and blocked for 1. **Depth** is one.

Training labels determined both **leaf** majorities and the error comparison that selected near rather than tall.

Without opening the old lesson, explain one key idea from Accuracy as a Fraction. Give an example and a mistake that the idea helps you avoid.

Look for: Explain why overall accuracy can hide failure on a less common class

Reference example, not the only acceptable wording: The **training**-majority rule always predicts clear. Matrix [[16,0],[4,0]] gives 16/20=80 percent, FP=0,FN=4.

It misses all four blocked cases, so the overall score hides complete failure on that evaluated class.

Without opening the old lesson, explain one key idea from Features and Distractors. Give an example and a mistake that the idea helps you avoid.

Look for: Choose recorded properties that preserve distinctions needed for a stated task

Reference example, not the only acceptable wording: Use length and width measured before the decision. State the order as length in cm, then width in cm; the record is 12, 8.

The target is fits or too large under the guide, and this card is fits because both equalities are included.

Colour and card number are not needed for this rule; the final inspection sticker copies the answer and is unavailable before inspection.

The cutoff was selected using **training** labels. Matching those examples does not establish performance on new examples; validation can expose mistakes and support a baseline comparison.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Decision Trees

Accuracy as a Fraction

Features and Distractors

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Six cards contain three animal and three space books; all have blue covers. Compare asking 'blue cover?' with 'animal subject?'. Is the better question guaranteed to work on inaccurately tagged books?

Concept to explain

Week 15: Choose **Informative** Questions

Term	Meaning
informative	Providing evidence that helps distinguish relevant possibilities or reduce uncertainty.
split	Separate a dataset into portions with different development or evaluation roles.
remaining possibilities	The options still consistent with the evidence after some have been ruled out.
efficiency	The resources required to achieve a result, such as time, memory or number of steps.

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach a worst-case question audit with a complete card table

> Keep the main identification criterion separate from retained tree-fitting review.

State the guessing game and candidate comparison

A fictional guessing game has eight catalogue cards and asks yes/no questions to narrow the target.

Cards: four mystery, four science; six paperback, two hardback; all English.

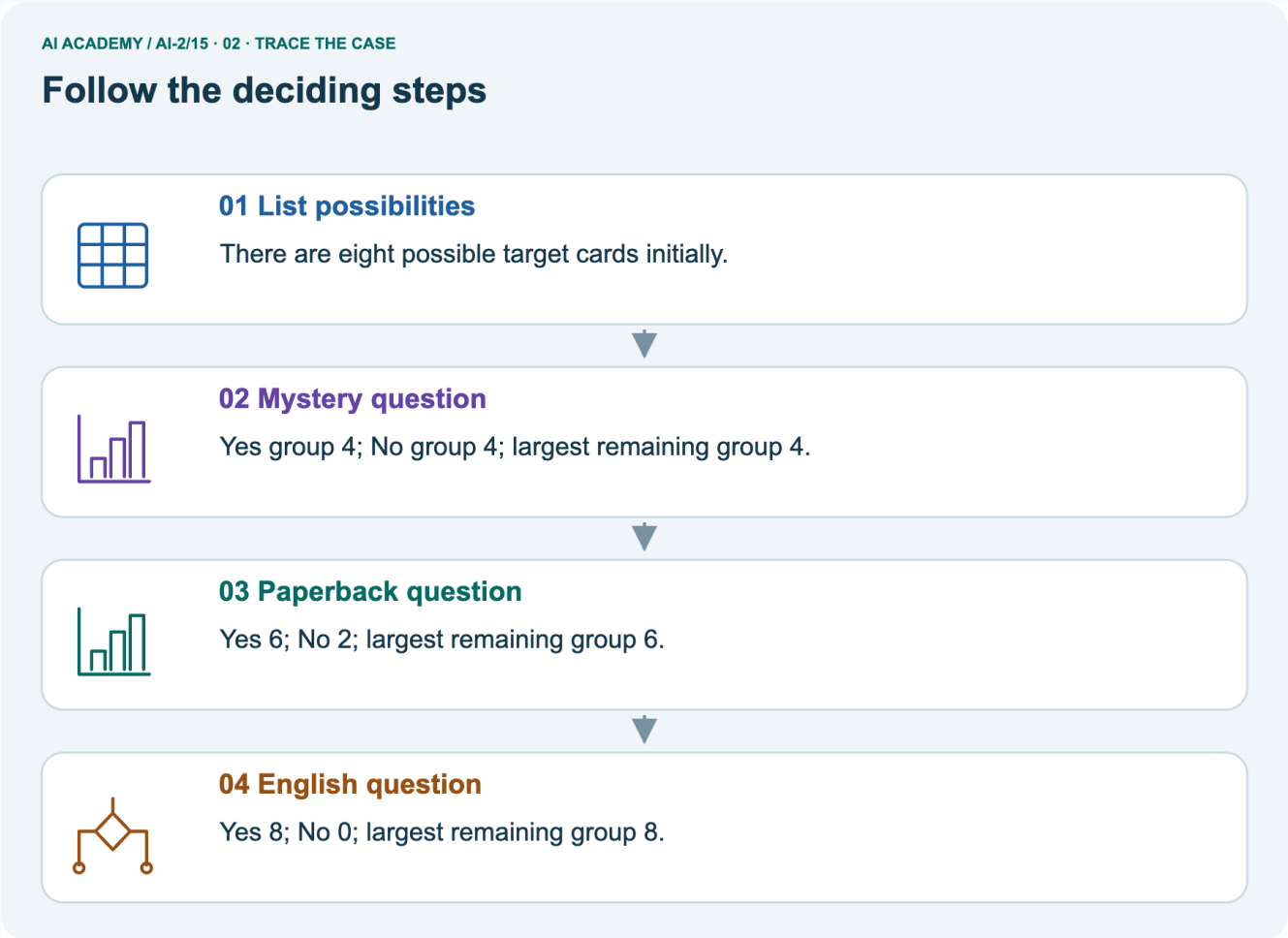
Compare worst-case remaining group size after asking mystery?, paperback? or English?. No claim about unequal target probabilities is made.

Supply the complete joint table

Card	Subject	Binding	Language
M1	mystery	paperback	English
M2	mystery	paperback	English
M3	mystery	paperback	English
M4	mystery	hardback	English
S1	science	paperback	English

Card	Subject	Binding	Language
S2	science	paperback	English
S3	science	paperback	English
S4	science	hardback	English

Trace the original comparison



Follow the labelled picture with the matching explanation. Open the vector diagram

Under the stated worst-case criterion, mystery is the most useful of these questions because it leaves at most four cards. The constant language clue never narrows the set.

This is an identification-game criterion; a classifier **fitting** labels may instead compare **prediction** errors or another declared **training** measure.

Predict the restricted four-card set

Change: The current possibility set contains only the four mystery cards.

Prediction: Asking mystery now leaves all four; it no longer narrows the set.

Reason: Question value depends on the remaining candidates, not just on its wording.

Run the complete question audit

```
cards = [("M1", "mystery", "paperback", "English"),
         ("M2", "mystery", "paperback", "English"),
         ("M3", "mystery", "paperback", "English"),
         ("M4", "mystery", "hardback", "English"),
         ("S1", "science", "paperback", "English"),
         ("S2", "science", "paperback", "English"),
         ("S3", "science", "paperback", "English"),
         ("S4", "science", "hardback", "English")]
questions = [("mystery?", 1, "mystery"),
             ("paperback?", 2, "paperback"),
             ("English?", 3, "English")]
def audit(possibilities):
    best = None
    best_worst = len(possibilities) + 1
    for name, column, value in questions:
        yes = sum(card[column] == value for card in possibilities)
        no = len(possibilities) - yes
        worst = max(yes, no)
        print(name, yes, no, worst)
        if worst < best_worst:
            best, best_worst = name, worst
    return best
print("chosen", audit(cards))
remaining = [card for card in cards if card[1] == "mystery"]
print("after mystery Yes", audit(remaining))
```

Discuss assumptions and elicit an explanation

Balanced groups do not guarantee a truthful user answer or optimal choices under every probability or cost model.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Week 14 review · Fitted-tree comparison

A decision tree asks a feature question and follows a **branch** to a **prediction**. A tree with one question has **depth** one and is sometimes called a stump. This retained Week 14 review fits a question and its **leaf prediction**s from training labels. The main Week 15 guessing-game task instead compares the largest group of remaining possible target cards.

Our six fictional **training** rows have two binary sensor flags available before **prediction**: near and tall. A value of 1 means that flag is present; 0 means absent. The reviewed scene label is separate.

Each row represents a different trip on a familiar route, including rows with identical values. Similar readings can have different scene labels. These are teaching examples, not robot control instructions.

Try near first and tall second. For each question, group the **training** rows by 0 and 1, then predict each group’s majority label; choose clear for tied label counts.

Count all **training** mistakes and keep the first question if scores tie. Stop after one question. This is our stated classroom **fitting** procedure.

Feature 0 is near; feature 1 is tall; row[2] is the label. **Output** is 0 1, then 1 2, then 0 clear blocked. Near makes one **training** mistake: near=0 contains two clear labels and one blocked, so that **leaf** predicts clear.

Near=1 contains three blocked labels. Tall makes two mistakes because both its groups have a blocked majority.

The stored tree contains the chosen feature and its two learned leaves. The code assumes nonempty data, binary flags, allowed labels and examples on both **branch**es of each candidate.

An empty **branch** would need a declared fallback; this small example does not implement one. The score is total **training** mistakes, not an average giving small and large **branch**es equal weight.

This is an explicitly limited **depth**-one learner. General tree algorithms can repeat **split** selection within **branch**es and often use impurity criteria such as Gini. Our code does not implement that full procedure.

Keeping the scope small lets you inspect every decision.

Two paths through the learned tree

ROOT QUESTION	BRANCH ANSWER	LEARNED LEAF
Is near equal to 1	No near is 0	Predict clear
Is near equal to 1	Yes near is 1	Predict blocked

Both paths start at the same root. Tall is not used by this fitted tree; the leaf labels came from training counts.

Validation rows are (near=1,tall=1,blocked), (0,0,clear), (0,1,blocked), (1,0,clear). Follow only near: **prediction**s are blocked, clear, clear, blocked. Two of four match.

The **training**-majority baseline predicts blocked, since **training** has four blocked and two clear; it also matches two of four validation labels.

> **Which validation rows does the tree miss, and which does the baseline miss? Explain why the tall value cannot change this tree's prediction.**

Keep the fitted **root** and leaves unchanged while scoring. Equal accuracy does not mean identical errors, and five correct **training prediction**s out of six is not the validation result.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Eight cards remain.

2. ‘Has wings?’ **split**s 4/4; worst **branch** leaves 4.

3. ‘Is red?’ **split**s 1/7; worst **branch** leaves 7.

4. For this set, wings is the more **informative** first question.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

Eight blue cards contain four circles and four triangles; target is shape.

Which question is more useful: blue or circle?

Exact explanation to model

Circle separates target groups. Blue puts every card in one **branch**. Usefulness depends on the task and examples.

Explain the added worked example

Key terms

Term	Meaning
Root	The first question asked by a tree

Term	Meaning
Branch	A path selected by the answer to a question
Leaf	An endpoint that stores a prediction
Depth	The largest number of question steps from root to leaf

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

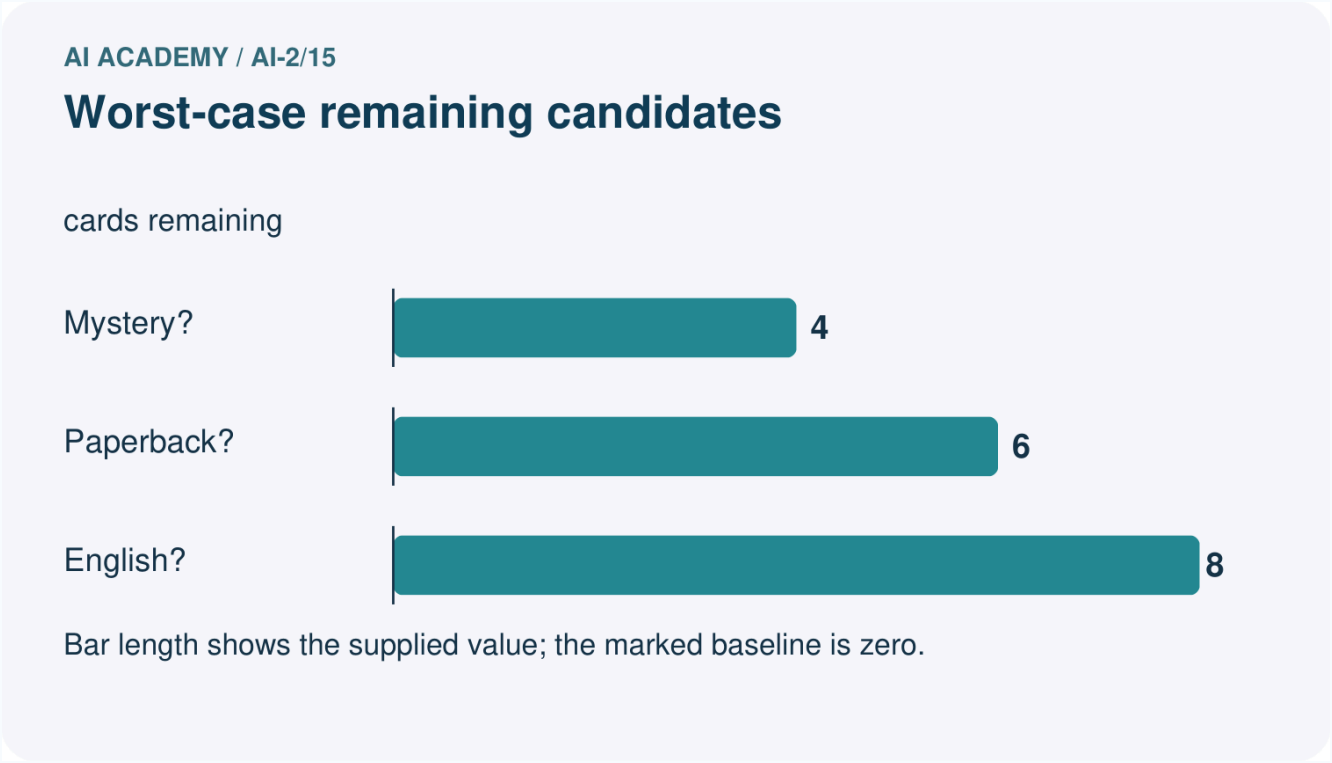
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Choose Informative Questions

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/15

Choose Informative Questions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. List possibilities

There are eight possible target cards initially.



2. Mystery question

Yes group 4; No group 4; largest remaining group 4.



3. Paperback question

Yes 6; No 2; largest remaining group 6.



4. English question

Yes 8; No 0; largest remaining group 8.

CHANGE ONE THING

The current possibility set contains only the four mystery cards.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. List possibilities	There are eight possible target cards initially.
2. Mystery question	Yes group 4; No group 4; largest remaining group 4.
3. Paperback question	Yes 6; No 2; largest remaining group 6.
4. English question	Yes 8; No 0; largest remaining group 8.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide the current question comparison, then fade help

> Collect each learner's branch counts before discussion.

Main sequence

Main sequence: read all eight cards and state the worst-case criterion; collect three split predictions; count each branch before revealing code; restrict to M1–M4 and recount; complete current W1–W3 with fading help; collect ten-card W4 independently; discuss current W5 assumptions and record a changed-case return. Schedule the retained tree material only as optional Week 14 review.

Current W2 prompt and checked reasoning

Completion task: For mystery?, paperback? and English?, count Yes and No, calculate the largest remaining group and select the best under the stated criterion. Explain why one useful question need not identify a unique target.

Teacher check: Mystery splits4/4, paperback6/2, English8/0, with worst remaining groups4,6,8. Choose mystery; a branch still contains four possible targets. Use the complete supplied card table and the declared criterion.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Challenge a repeated constant clue

Proposed explanation: The best first question stays informative after its answer is already known. Test this on the four remaining mystery cards and show both **branch** sizes.

Expected correction: mystery is now4/0 and leaves all four; paperback is3/1 and leaves at most three on the supplied joint table.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Use the pictured comparison

Choose the best original question and explain why its more balanced **split** matters here.

Mystery gives 4/4 and worst case 4, better than paperback's 6 and English's 8 under the declared smallest-worst-case rule.

Changed-case reasoning: Asking mystery now leaves all four; it no longer narrows the set. Question value depends on the remaining candidates, not just on its wording.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Week 14 review · Earlier tree teaching sequence

Guided game counts: original mystery4/4, paperback6/2, English8/0; choose mystery by largest groups4,6,8. After mystery Yes, M1–M4 yield4/0,3/1,4/0; choose paperback by largest groups4,3,4. Neither chosen answer necessarily identifies a unique target. Keep identification-game counts distinct from the earlier fitted-tree validation scores.

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve training search versus validation. Introduce root, branch, leaf and depth with one question.
7 to 18 minutes	Sort the six cards by near. Count labels within each branch and assign its majority without using validation.
18 to 32 minutes	Repeat for tall and complete W1 and W2. Add branch mistakes to score the full training set.
32 to 45 minutes	Trace code and the diagram. Predict all four validation rows, complete the guided question and compare the baseline.

Stage	Teacher action
Session 2 start to 8 minutes	Retrieve why a leaf label is learned rather than dictated by the feature name.
8 to 20 minutes	Complete W3 and discuss depth limits, tied questions and the assumptions required by the short code.
20 to 35 minutes	Collect W4 independently before hints. Require branch counts, both candidate scores, learned leaves and validation comparison.
35 to 45 minutes	Use W5 and the exit claim. Give the fresh retry after feedback and record support.

▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

Board sequence

Write the given facts: Eight blue cards contain four circles and four triangles; target is shape.

Write the question to explain: Which question is more useful: blue or circle?

Reveal after **prediction**s: Circle separates target groups. Blue puts every card in one **branch**. Usefulness depends on the task and examples.

Use the earlier changed question as optional later practice. Reserve Changed independent case for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Case-specific teaching sequence

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied card-set audit

> Keep the criterion and current set beside every count.

Current investigation and answer guidance

Original eight-card game: mystery4/4→worst4; paperback6/2→6; English8/0→8. Choose mystery under smallest-worst-case. Restricted M1–M4: mystery4/0→4, paperback3/1→3, English4/0→4; choose paperback. One answer does not uniquely identify every card. The supplied joint table supports the changed binding count; the marginals alone would not. Tree-fitting excerpts are retained Week 14 review, not the current W1–W3/W5. Reserve ten-card pet/space W4 for independent work.

Predict, count and recount

1. Predict → Test → Record → Explain

2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.

3. Use the complete eight-card catalogue table. Count both branches and their maximum for each question, select by the declared worst-case rule, then restrict to four mystery cards and recount. Preserve the current population and criteria beside each result.

4. What did the hardest test teach you that the normal test did not?

Keep the evidence

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the current misconceptions

Current misconceptions: a constant question splits into two nonempty groups—it does not, English gives8/0. The best original question stays best after an answer—recount; mystery becomes4/0. A smaller worst branch identifies the target—four or three candidates can still remain. Worst-case count gives the minimum expected cost—not without a probability/cost model. A fabricated answer is evidence—record only supplied assumptions or labelled simulated answers.

Choose access support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Week 14 review · Earlier tree misconceptions

A near **branch** must predict blocked: The **branch** label is learned from its **training** labels; W4 deliberately reverses that relationship.

Every feature is used during **prediction**: Only questions in the stored tree are followed. Candidate questions rejected during **fitting** do not affect this tree's **output**.

A deeper tree must work better: More questions can fit **training** detail. Future performance needs separate evaluation; **depth** choices must be documented.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Changed independent case for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected ten-card comparison and current game tasks

> Keep the selected W4 answer until after the first attempt.

Selected independent question and key

Selected case: Ten cards have six pet subjects and four space subjects; all are green. Compare asking “green?” with asking “pet?” to identify a card. Give the largest remaining group and one limitation.

Green **split**s 10/0, leaving as many as 10. Pet **split**s 6/4, leaving as many as 6, so it is more **informative** under these supplied counts.

Incorrect subject tags can invalidate the **split**; one question does not uniquely identify every card.

Judge reasoning and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.

Dimension	Evidence required
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give an actionable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Current W1–W3 and W5 answer guidance

Current W1: M1–M4,S1–S4 are eight possible targets; truthful answers retain the matching branch, English is8/0. W2: mystery4/4 worst4, paperback6/2 worst6, English8/0 worst8; choose mystery but four candidates can remain. W3: M1–M4 give mystery4/0 worst4, paperback3/1 worst3, English4/0 worst4; choose paperback. W5: truthful answers, correct complete tags and the specified candidate set/criterion are assumptions. Expected cost needs probabilities and costs; tie order is reproducibility, not a claim of superior future usefulness.

▼ Week 14 review · Earlier tree-task keys

W1 Name the parts and decisions

Identify the **root**, its two **branch**es, **leaf prediction**s and **depth** of the worked fitted tree. Which parts were selected using **training** labels?

The **root** asks near, with **branch**es 0 and 1. Leaves predict clear for 0 and blocked for 1. **Depth** is one. **Training** labels determined both **leaf** majorities and the error comparison that selected near rather than tall.

W2 Verify the candidate scores

For each near **branch** and each tall **branch**, count blocked and clear labels and choose a **leaf**. Give the total errors for each candidate question.

Near=0 has 1 blocked and 2 clear, predicts clear with 1 error; near=1 has 3 blocked and 0 clear, predicts blocked with 0 errors. Tall=0 has 2 blocked and 1 clear; tall=1 also has 2 blocked and 1 clear.

Both tall leaves predict blocked with 1 error each. Totals are near 1 and tall 2.

W3 Separate fitting from prediction

A learner sees a blocked validation case with near=0 and changes that **leaf** to blocked before reporting the score. Explain what changed and how to report the original fitted tree fairly.

The **leaf** was changed using validation **evidence**, so it is no longer the original **training**-fitted tree. Score the original stored leaves on all the same validation rows and record any later development change separately.

Preserve a separate final test for final evaluation.

W4 Fit a new tree independently

New **training** rows (near,tall,label) are (1,0,clear),(1,1,clear),(1,0,clear),(0,1,blocked),(0,0,blocked),(0,1,clear). Use the same candidate order and tie policies. Give both candidates' leaves and errors, then select the tree.

Validation rows are (1,1,clear),(0,0,blocked),(0,1,clear),(1,0,blocked). List tree **prediction**s and compare accuracy with the **training**-majority baseline.

Near=0 predicts blocked from 2 blocked and 1 clear; near=1 predicts clear from 3 clear. Near has 1 error. Both tall groups have 2 clear and 1 blocked, so both leaves predict clear and total errors are 2.

Choose near with reversed **leaf** labels compared with the worked example. Validation **prediction**s clear,blocked,blocked,clear match rows 1,2: 2/4. Baseline clear matches rows 1,3: 2/4. Both are 50 percent but errors differ.

W5 Audit a tie and a claim

If near and tall make the same **training** error count, which question does this code retain and why? Does retaining that question prove it will be better on future trips?

It retains near because near is tried first and strict < replaces only for fewer errors. This is a declared tie convention, not **evidence** of better future performance. Separate evaluation and relevant examples are still needed.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Circle separates target groups. Blue puts every card in one **branch**. Usefulness depends on the task and examples.

Additional transfer practice question

Cards are two cats and two dogs, all red. Which question helps separate species: is it red or does it bark? State your assumption.

Reasoned practice answer

Is it red separates nothing here. Does it bark could separate the example categories if the supplied facts consistently distinguish them; real behaviour is not a perfect species rule.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Any question creates useful separation.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Blue sends all six to one **branch** and separates nothing. Animal subject separates the supplied labels into two groups of three.

Its usefulness depends on accurate subject tags and the intended target; new data can expose missing or misleading tags.

Explanations for the additional workbook examples

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

Ask first about preferences that remove the largest number of unsuitable books.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Choose recommender questions that distinguish feasible options, documenting the decision criterion rather than asking unnecessary personal details.

Support while preserving the reasoning goal

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Physically move the current cards into labelled Yes/No areas. Write both counts and circle the larger. After an answer, remove excluded cards before recounting. Use written field values so the task does not depend on colour vision. Extension: explain why worst-case count and probability-weighted expected cost are different questions.

For selected W4, green gives10/0 and worst10; pet gives6/4 and worst6, so choose pet under this criterion. Require the limitation that a question need not identify one card and tags or answers can be wrong. Fresh retry: nine cards, all striped, with five sea subjects and four land; striped9/0→9, sea5/4→5. Record a first independent explanation before feedback.

Plan the delayed check

Suggested return points in this level: ai-2/16, ai-2/18, ai-2/21. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 16: Over**fitting** Intuition. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week 16: Overfitting Intuition. Carry forward the distinction between a procedure's declared objective and evidence that it works beyond the examples used to choose it. Retained fitted-tree library references concern Week 14 review rather than a library implementation of this guessing game.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Transparent Book Recommender

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: Compare questions by their largest remaining group

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Practice and assessment reminders

Before class, solve the supported practice: For each near **branch** and each tall **branch**, count blocked and clear labels and choose a **leaf**. Give the total errors for each candidate question.

Reasoned practice answer: Near=0 has 1 blocked and 2 clear, predicts clear with 1 error; near=1 has 3 blocked and 0 clear, predicts blocked with 0 errors.

Tall=0 has 2 blocked and 1 clear; tall=1 also has 2 blocked and 1 clear. Both tall leaves predict blocked with 1 error each. Totals are near 1 and tall 2.

Check every learner before discussion: Use the worked case for **Fitting** a decision tree. Proposed explanation: “A near **branch** must predict blocked”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: The **branch** label is learned from its **training** labels; W4 deliberately reverses that relationship.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: The current possibility set contains only the four mystery cards. Require a **prediction** and a reason before comparing. Expected result: Asking mystery now leaves all four; it no longer narrows the set.

Reason: Question value depends on the remaining candidates, not just on its wording.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ Earlier wording, alternatives and picture transcripts

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The **branch**-boundary and indexing lessons deliberately include failure cases.

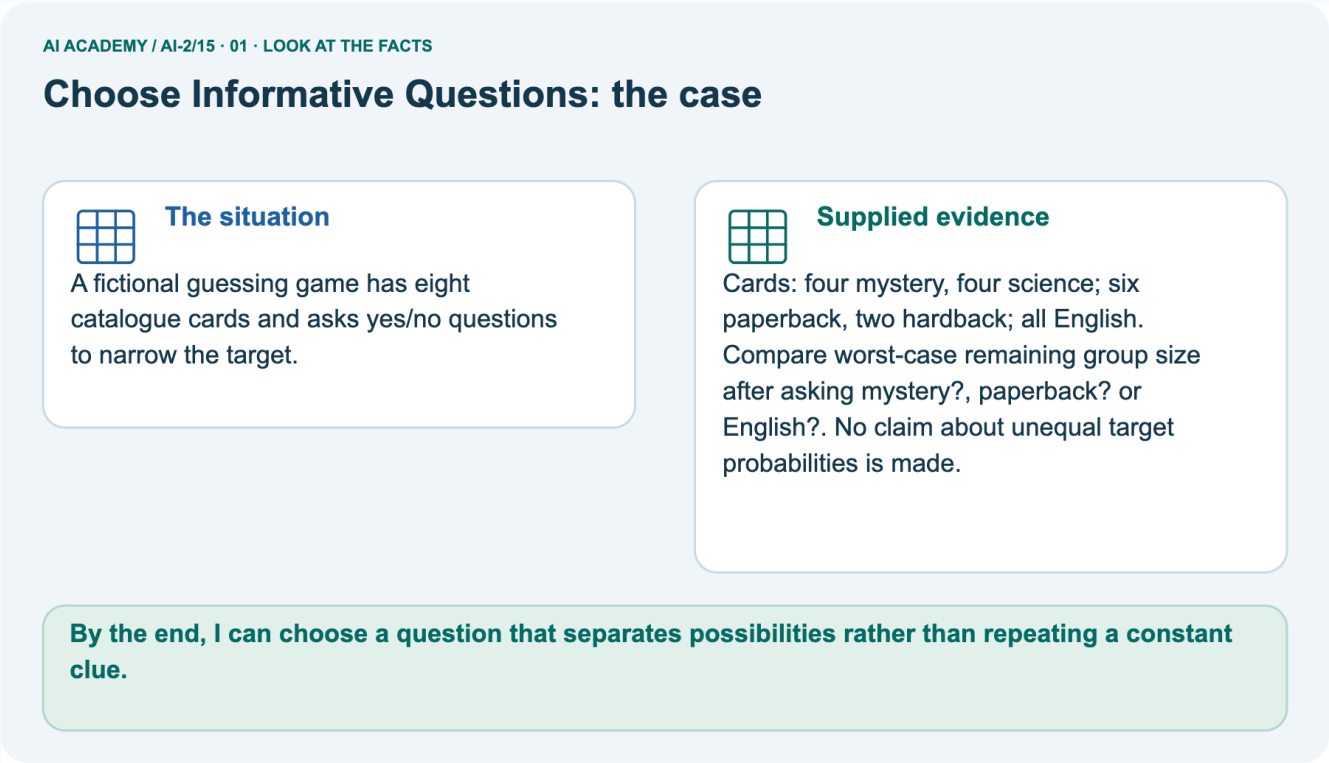
▼ Reference: teach with the six pictures

REFERENCE · SECTION 9

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Choose Informative Questions: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

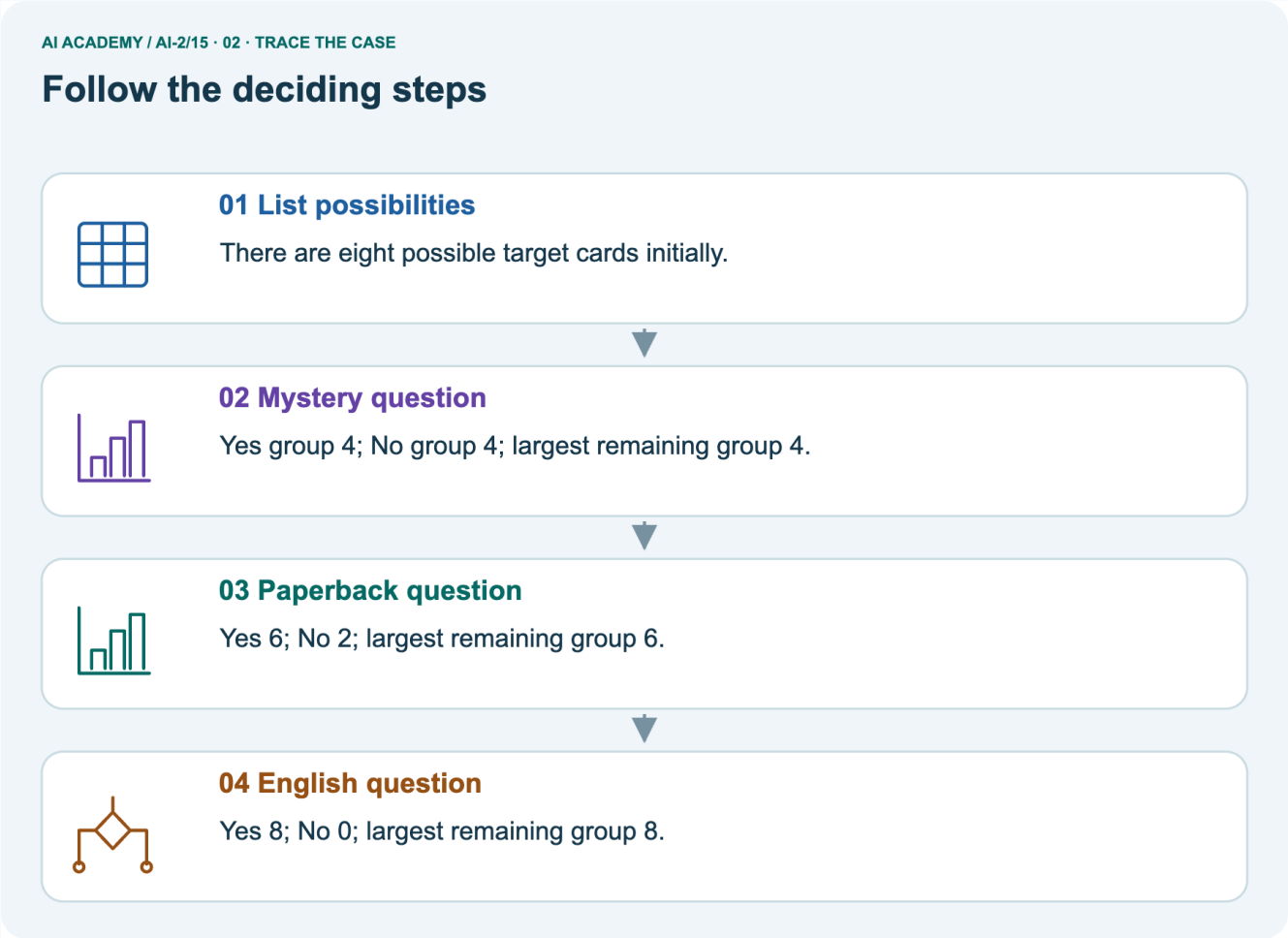
A fictional guessing game has eight catalogue cards and asks yes/no questions to narrow the target. Supplied facts: Cards: four mystery, four science; six paperback, two hardback; all English.

Compare worst-case remaining group size after asking mystery?, paperback? or English?. No claim about unequal target probabilities is made.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



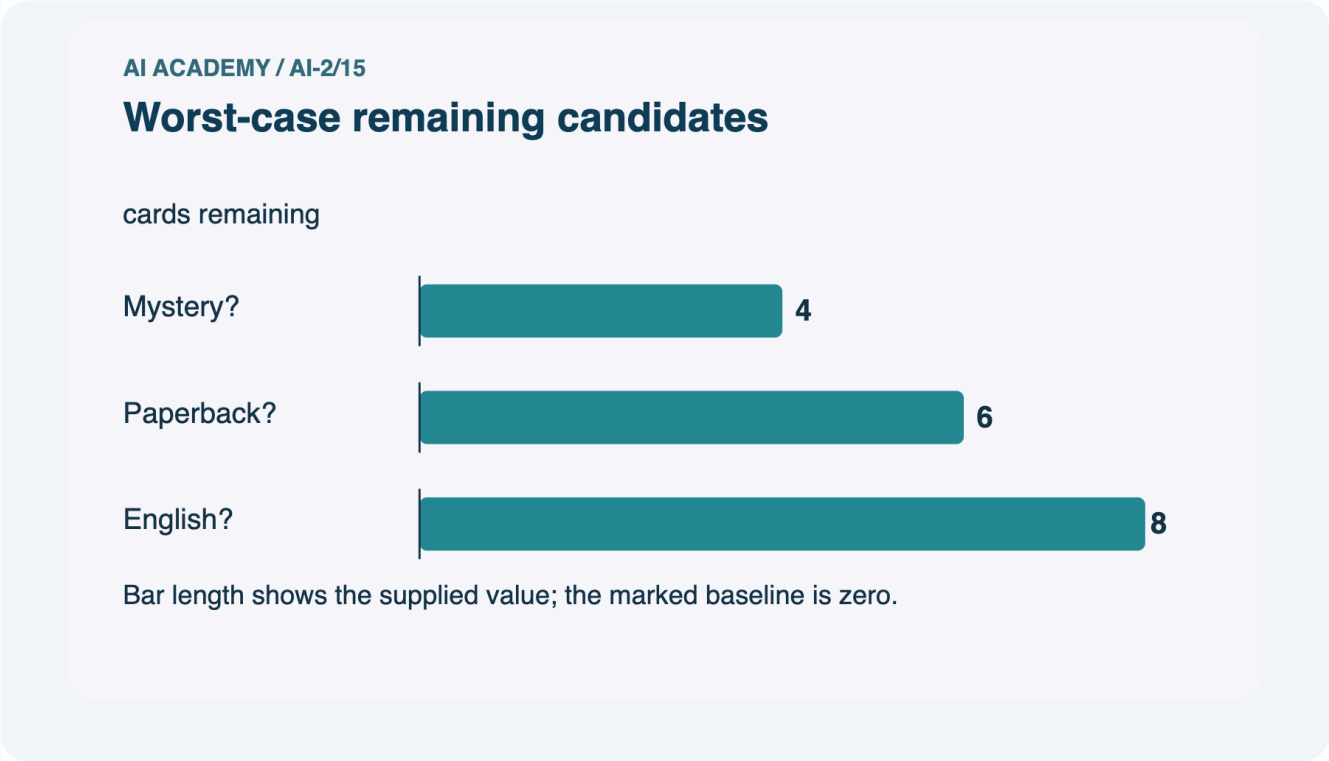
Follow the labelled picture with the matching explanation. Open the vector diagram

List possibilities: There are eight possible target cards initially. → Mystery question: Yes group 4; No group 4; largest remaining group 4. → Paperback question: Yes 6; No 2; largest remaining group 6. → English question: Yes 8; No 0; largest remaining group 8.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Worst-case remaining candidates



Follow the labelled picture with the matching explanation. Open the vector diagram

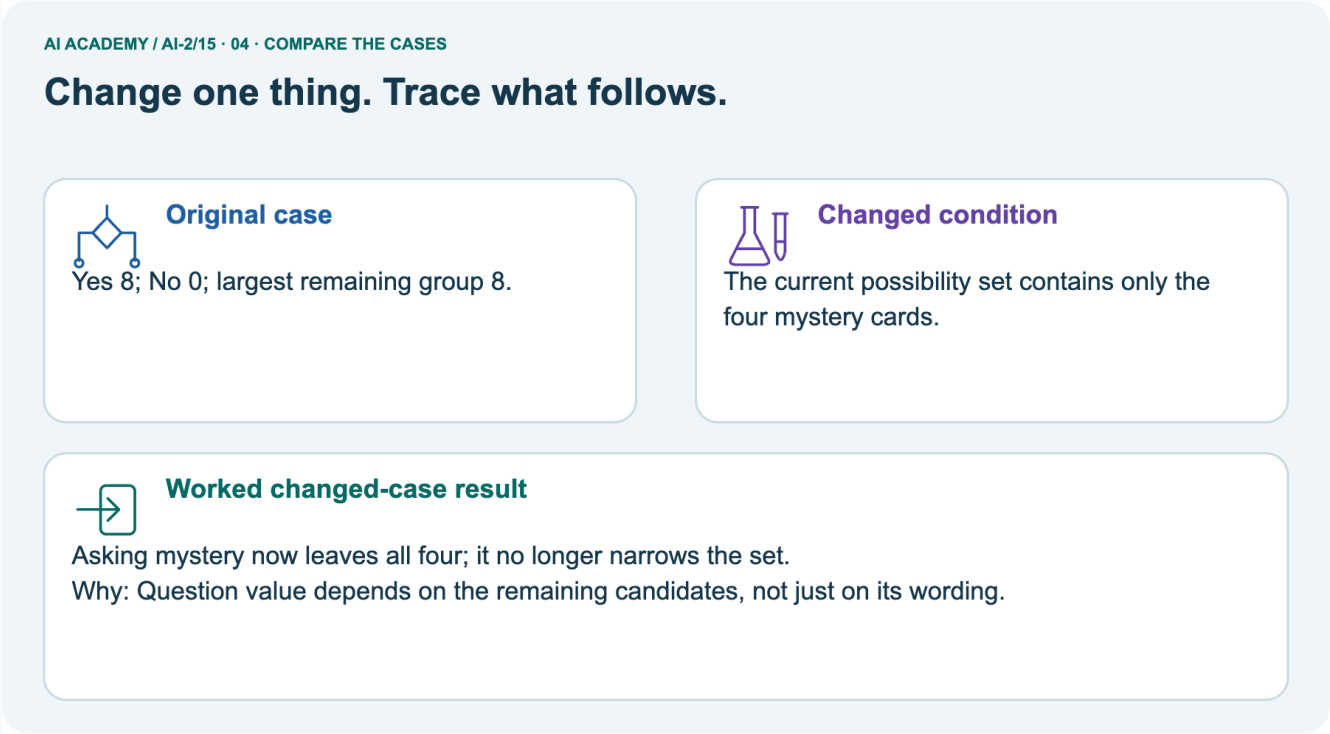
Worst-case remaining candidates. Units: cards remaining. Mystery?: 4. Paperback?: 6. English?: 8.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Yes 8; No 0; largest remaining group 8. Changed condition: The current possibility set contains only the four mystery cards. Expected result: Asking mystery now leaves all four; it no longer narrows the set.

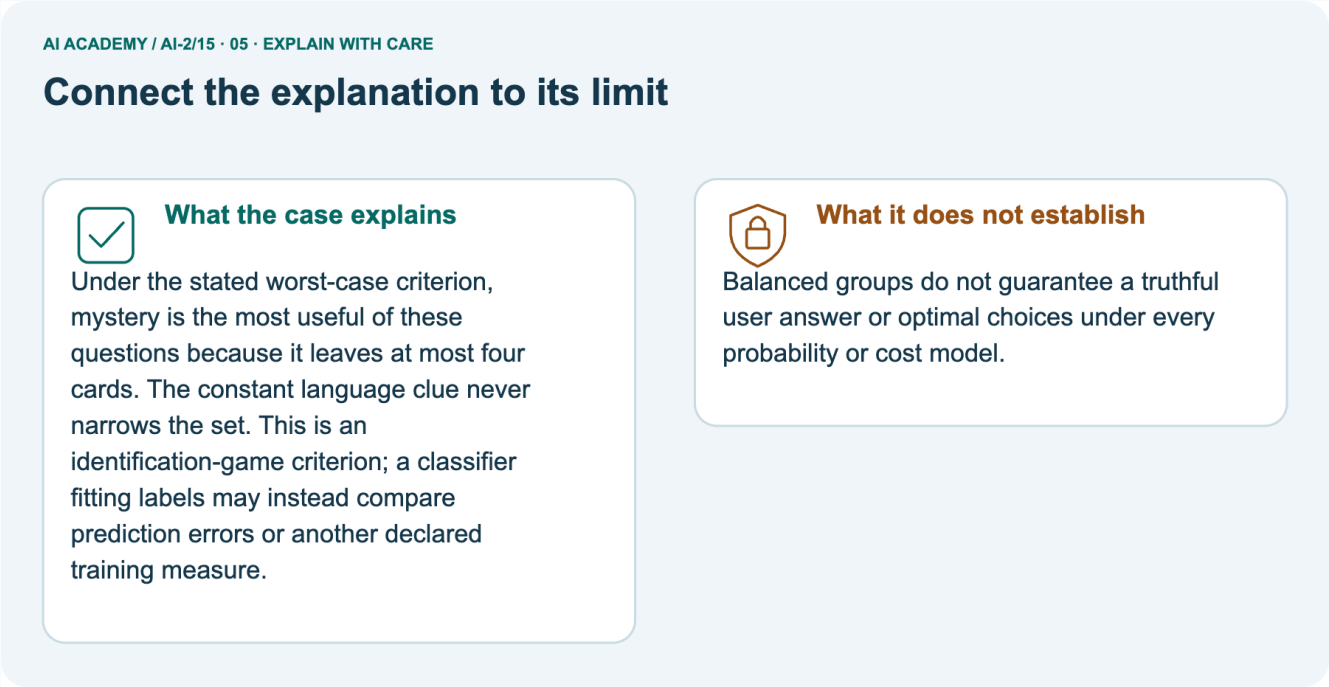
Reason: Question value depends on the remaining candidates, not just on its wording.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Question value depends on the remaining candidates, not just on its wording.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Under the stated worst-case criterion, mystery is the most useful of these questions because it leaves at most four cards. The constant language clue never narrows the set.

This is an identification-game criterion; a classifier **fitting** labels may instead compare **prediction** errors or another declared **training** measure.

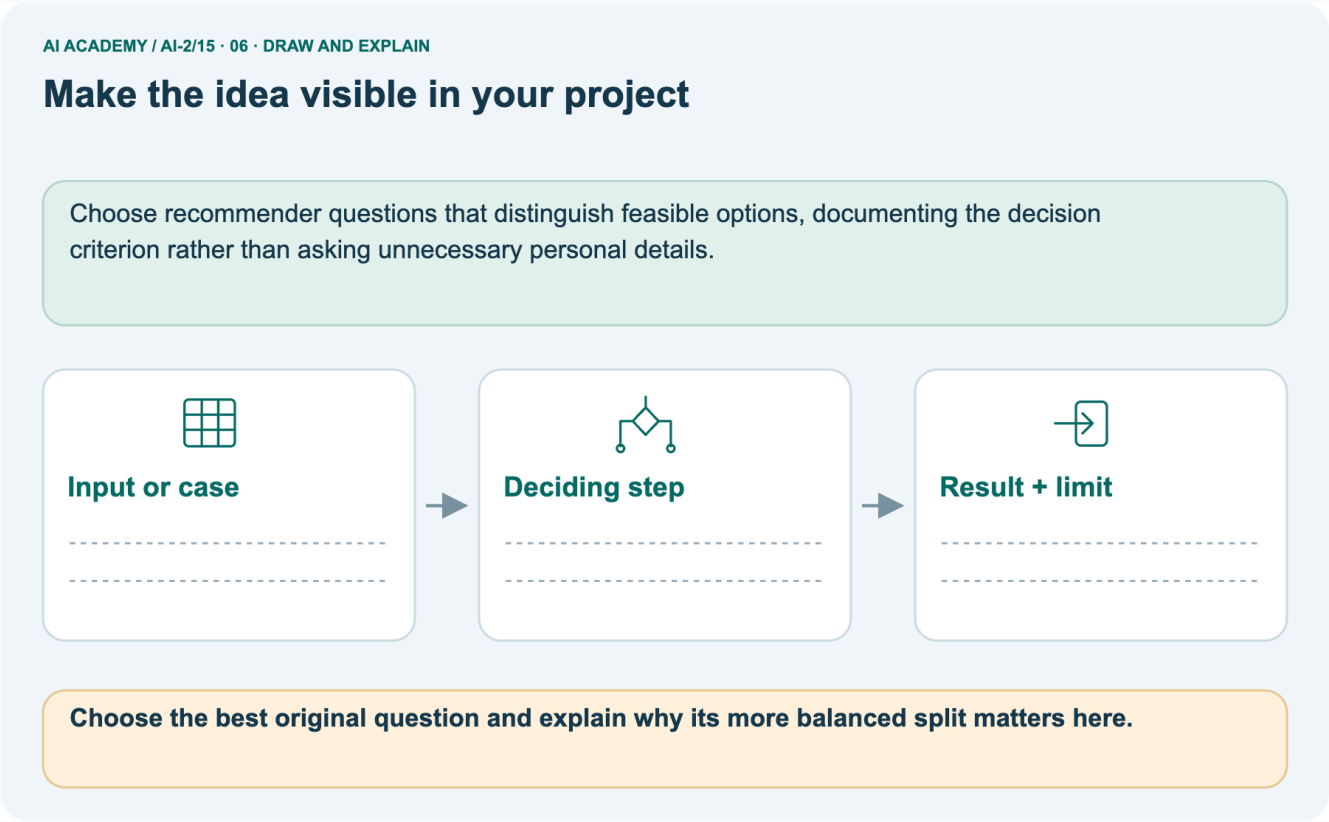
Limit: Balanced groups do not guarantee a truthful user answer or optimal choices under every probability or cost model.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Balanced groups do not guarantee a truthful user answer or optimal choices under every probability or cost model.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Choose recommender questions that distinguish feasible options, documenting the decision criterion rather than asking unnecessary personal details. Draw your **input**, deciding step, result and limit.

Supported question: Choose the best original question and explain why its more balanced **split** matters here.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Mystery gives 4/4 and worst case 4, better than paperback's 6 and English's 8 under the declared smallest-worst-case rule.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Choose Informative Questions: the case

02. Follow the deciding steps

03. Worst-case remaining candidates

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.